

SHENNONG OBJECT-CENTERED WORKFLOW

输入 -> 对象 -> 分析模块 -> versioned result -> 可视化与交付

Method registry 方法发现

sn_list_methods(task) | sn_get_method_status(method, task)
23 tasks; runtime, availability, defaults, citations

Input 输入

10x / STARsolo / H5AD / qs2
matrix / list / SummarizedExperiment
sn_read()
sn_list_10x_paths()

Prepare & QC 预处理

sn_initialize_seurat_object()
sn_add_qc_metrics()
sn_filter_cells() / sn_filter_genes()
sn_normalize_data()
sn_find_doublets()
sn_remove_ambient_contamination()

Shennong object

Seurat-centered mutable analysis object
Assays + layers: RNA / ADT / spatial
meta.data: cell + sample provenance
reductions / graphs / images
misc\$shennong\$results[type][result_id]
commands + reproducibility provenance

Standalone bulk lane

matrix / list / SummarizedExperiment
sn_run_bulk(workflow=)
sn_assess_bulk_qc() / sn_find_de()
sn_score_bulk_pathways() / sn_run_wgcna()
sn_run_survival() / sn_run_clinical_association()

Analysis modules 分析模块

Each module consumes the object or a validated standalone input.

Cluster & Integrate

sn_run_cluster()
sn_run_multimodal()
Harmony / CCA / RPCA
scVI / scANVI / scPoli
WNN / totalVI / Coralysis

Annotation, DE & Pathway

sn_run_annotation()
sn_find_de()
sn_run_enrichment()
review confidence + ontology

Programs & Regulation

sn_score_programs()
sn_test_programs()
sn_discover_programs()
sn_run_grn()
sn_run_metabolism()

Composition, DA & CNV

sn_calculate_composition()
sn_compare_composition()
sn_test_abundance()
sn_prioritize_states()
sn_run_cnv()

Communication & Dynamics

sn_run_cell_communication()
sn_run_trajectory()
sn_run_velocity()
sn_run_fate()

Spatial

sn_run_spatial(task=)
SVG / domain
neighborhood / mapping
deconvolution / communication

Versioned result 结果契约

schema_version = 2.0.0
analysis_type + result_id
tables\$primary + diagnostics
provenance + backend
sn_list_results() / sn_get_result()
sn_store_result() / sn_validate_result()

Visualization 可视化

sn_list_plot_methods()
26 analysis types / 78 views
sn_plot_result(object,
analysis_type, result_id, view)
direct plots: dim / feature / composition
distribution / association

Validate & Export 输出

sn_apply_figure_profile()
sn_validate_figure()
sn_export_figure()
sn_export_figure_bundle()